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1. Concept Selection:

1.1 LED Configuration:

$L_{strip} := 1\ m$

$N_{strip} := 60 \cdot \left(\frac{LED}{m} \right)$

$N_{strips} := 2 \cdot \frac{1}{LED}$

$N_{LED} := L_{strip} \cdot N_{strip} \cdot N_{strips} = 120$ //Total number of LEDs

1.2 WS2812B Data Requirement

$B_{LED} := 24$

$f_{data} := 800\ kHz$

$t_{LED} := \frac{B_{LED}}{f_{data}} = 3 \cdot 10^{-5}\ s$ //The time required to transmit the data for one LED

1.3 Time Required for the Complete LED Strip

$B_{Total} := N_{LED} \cdot B_{LED} = 2880$ //The total number of bits transmitted to update all 120 LEDs

$t_{update} := \frac{B_{Total}}{f_{data}} = 0,0036\ s$ //The total transmission time

$t_{reset} := 0,05\ ms$ //WS2812B minimum reset (latch) time from the datasheet

$t_{strip} := t_{update} + t_{reset} = 3,65\ ms$ //Complete strip update including the reset time

1.4 Required Lighting Update Rate

$f_{required} := 50\ Hz$ //Assumption for the comparison

$T_{required} := \frac{1}{f_{required}} = 0,02\ s$

$U_{LED} := \left(\frac{t_{update}}{T_{required}} \right) \cdot 100 = 18\%$ //The percentage of the available update period used by the LED data transmission

$t_{remaining} := T_{required} - t_{update} = 0,0164\ s$

Conclusion: This means the LED data transmission occupies 18% of the 20 ms update period, leaving approximately 16.4 ms for other processing.:

1.5 Maximum Theoretical LED Update Rate

$f_{max_LED} := \frac{1}{t_{update}} = 277,7778\ Hz$

1.6 Concept 1 - Raspberry Pi Pico 2 (RP2350)

1.6.1 Microcontroller processing speed:

$f_{CPU_1} := 150\ MHz$

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$T_{CPU_1} := \frac{1}{f_{CPU_1}} = 6,6667 \cdot 10^{-9}\ s$

1.6.2 Available Processing Cycles per Lighting Update:

$C_{Total_1} := f_{CPU_1} \cdot T_{required} = 3 \cdot 10^{6}\ cycles$

1.6.3 Processing Cycles During LED Transmission:

$C_{LED_1} := f_{CPU_1} \cdot t_{update} = 5,4 \cdot 10^{5}\ cycles$

$C_{remaining_1} := C_{Total_1} - C_{LED_1} = 2,46 \cdot 10^{6}\ cycles$

1.7 Concept 2 — Arduino Nano

1.7.1 Microcontroller Processing Speed:

$f_{CPU_2} := 16\ MHz$

$T_{CPU_2} := \frac{1}{f_{CPU_2}} = 6,25 \cdot 10^{-8}\ s$

1.7.2 Available Processing Cycles per Lighting Update:

$C_{Total_2} := f_{CPU_2} \cdot T_{required} = 3,2 \cdot 10^{5}\ cycles$

1.7.3 Processing Cycles During LED Transmission:

$C_{LED_2} := f_{CPU_2} \cdot t_{update} = 57600\ cycles$

$C_{remaining_2} := C_{Total_2} - C_{LED_2} = 2,624 \cdot 10^{5}\ cycles$

1.8 Concept 3 — ESP32

1.8.1 Microcontroller Processing Speed:

$f_{CPU_3} := 240\ MHz$

$T_{CPU_3} := \frac{1}{f_{CPU_3}} = 4,1667 \cdot 10^{-9}\ s$

1.8.2 Available Processing Cycles per Lighting Update:

$C_{Total_3} := f_{CPU_3} \cdot T_{required} = 4,8 \cdot 10^{6}\ cycles$

1.8.3 Processing Cycles During LED Transmission:

$C_{LED_3} := f_{CPU_3} \cdot t_{update} = 8,64 \cdot 10^{5}\ cycles$

$C_{remaining_3} := C_{Total_3} - C_{LED_3} = 3,936 \cdot 10^{6}\ cycles$

Conclusion: For the three concepts, the LED requirement is identical because the same 120 WS2812B LEDs are used. The calculated transmission time is therefore 3.60 ms for all three concepts, which is below the 20 ms period associated with the 50 Hz lighting target. The main difference is the processing capacity available around the LED update. The Arduino Nano has approximately 320,000 clock cycles available per update period, compared with 3.00 million for the Pico 2 and 4.80 million for the ESP32.

1.9 MPU6050 I²C Data Transfer Time:

$N_{bytes} := 12$ //Six 16-bit accelerometer and gyroscope readings

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$f_{I2C} := 400\ kHz$

$t_{IMU} := \frac{N_{bytes} \cdot 8}{f_{I2C}} = 0,24\ ms$

Conclusion: The sensor data transfer time is 0.24 ms, which is well below the 100 ms motion detection target. Communication overhead and software processing add to this, but leave a large margin.

2. Detailed Design:

2.1 Power System Design

2.1.1 Battery Configuration:

$N_{battery} := 2$

$V_{cell,nom} := 3,6\ V$ $V_{cell,max} := 4,2\ V$

$C_{cell} := 3,4\ A\ hr$

$V_{battery,nom} := N_{battery} \cdot V_{cell,nom} = 7,2\ V$ $V_{battery,max} := N_{battery} \cdot V_{cell,max} = 8,4\ V$

$C_{battery} := C_{cell}$ //Cells are connected in series

2.1.2 Battery Energy:

$E_{battery} := V_{battery,nom} \cdot C_{battery} = 24,48\ W\ hr$

2.1.3 LED Power:

$I_{LED,max} := 60\ mA$

$I_{LED,total} := N_{LED} \cdot I_{LED,max} = 7,2\ A$

$V_{LED} := 5\ V$

$P_{LED} := V_{LED} \cdot I_{LED,total} = 36\ W$

2.1.4 Loads power:

$V_5 := 5\ V$

MAX98357A

$I_{amp} := 0,6\ A$

$P_{amp} := V_5 \cdot I_{amp} = 3\ W$

Raspberry Pi Pico 2

$I_{pico} := 0,1\ A$

$P_{pico} := V_5 \cdot I_{pico} = 0,5\ W$

Rotary encoder

$I_{encoder} := 0,01\ A$

$P_{encoder} := V_5 \cdot I_{encoder} = 0,05\ W$

Total 5 V power

$P_{5V} := P_{LED} + P_{amp} + P_{pico} + P_{encoder} = 39,55\ W$

$V_{3,3V} := 3,3\ V$

MPU6050

$I_{MPU} := 0,005\ A$

$P_{MPU} := V_{3,3V} \cdot I_{MPU} = 0,0165\ W$

OLED

$I_{OLED} := 0,03\ A$

$P_{OLED} := V_{3,3V} \cdot I_{OLED} = 0,099\ W$

MicroSD

$I_{SD} := 0,08\ A$

$P_{SD} := V_{3,3V} \cdot I_{SD} = 0,264\ W$

Total 3.3 V power

$P_{3,3V} := P_{MPU} + P_{OLED} + P_{SD} = 0,3795\ W$

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Total Theoretical System Power

$P_{total} := P_{5V} + P_{3,3V} = 39,9295\ W$

$I_{5V,eq} := \frac{P_{total}}{V_5} = 7,9859\ A$ //The theoretical worst-case load exceeds the 5 A rating of the selected buck converter.

$I_{3,3V,req} := \frac{P_{3,3V}}{V_{3,3V}} = 0,115\ A$ //Current drawn by the 3.3 V loads (MPU6050, OLED, MicroSD)

2.1.5 Power Demand at 58% LED Brightness:

Buck Converter Rating

$I_{converter,rated} := 5\ A$

$I_{margin} := I_{converter,rated} - I_{5V,eq} = -2,9859\ A$

Conclusion: This means the converter cannot support the theoretical maximum load with all 120 LEDs at full brightness and the brightness needs to be limited in software to overcome this constraint.

$Brightness := 0,58$

$I_{LED,58} := I_{LED,total} \cdot Brightness = 4,176\ A$

$P_{LED,58} := V_{LED} \cdot I_{LED,58} = 20,88\ W$

$P_{total,58} := P_{LED,58} + P_{amp} + P_{pico} + P_{encoder} + P_{3,3V} = 24,8095\ W$

$I_{5V,58} := \frac{P_{total,58}}{V_5} = 4,9619\ A$

$I_{margin,58} := I_{converter,rated} - I_{5V,58} = 0,0381\ A$

Conclusion: 58% is the highest brightness that keeps the theoretical 5 V load within the converter's 5 A rating, leaving a margin of only about 0.04 A. It is therefore treated as an upper brightness limit set in software. The measured maximum current below (2.91 A, safety factor ≈ 1.72) shows a much larger margin in practice, because not all LEDs run at full white simultaneously.

Measured Converter Current

$I_{measured,max} := 2,91\ A$

$P_{measured,max} := V_5 \cdot I_{measured,max} = 14,55\ W$

$I_{margin,measured} := I_{converter,rated} - I_{measured,max} = 2,09\ A$

$SF_{converter} := \frac{I_{converter,rated}}{I_{measured,max}} = 1,7182$

Buck Converter Duty Cycle

$V_{cell,min} := 3\ V$ //Li-ion discharge cut-off voltage per cell

$V_{battery,min} := N_{battery} \cdot V_{cell,min} = 6\ V$

$D_{nom} := \frac{V_5}{V_{battery,nom}} = 0,6944$

$D_{full} := \frac{V_5}{V_{battery,max}} = 0,5952$ //Fully charged battery

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$D_{empty} := \frac{V_5}{V_{battery,min}} = 0,8333$ //Battery at cut-off

Conclusion: The duty cycle varies between about 60% and 83% over the battery voltage range, which is within the normal operating range of a step-down converter.

2.1.6 Battery Calculations:

Converter Input Power:

$\eta_{converter} := 0,9$ //Assuming a converter efficiency of 90%

$P_{in} := \frac{P_{total}}{\eta_{converter}} = 44,3661\ W$ //For the theoretical maximum load

$P_{loss} := P_{in} - P_{total} = 4,4366\ W$ //Heat produced in the converter at the theoretical maximum load

$I_{battery} := \frac{P_{in}}{V_{battery,nom}} = 6,162\ A$

Battery C-Rate

$C_{rate} := \frac{I_{battery}\ hr}{C_{battery}} = 1,8123$ //Check against the cell's maximum continuous discharge rating

Cell and BMS Current Rating

$I_{cell,dis,max} := 8\ A$ //Samsung INR18650-35E maximum continuous discharge current (datasheet)

$SF_{cell} := \frac{I_{cell,dis,max}}{I_{battery}} = 1,2983$

$I_{BMS,rated} := 13\ A$ //HX-2S-D20 continuous current rating

$SF_{BMS} := \frac{I_{BMS,rated}}{I_{battery}} = 2,1097$

Usable Battery Capacity

$C_{usable} := 0,8 \cdot C_{battery} = 2,72\ A\ hr$ //Using an 80% usable capacity assumption

Theoretical Operating Time

$t_{operating} := \frac{C_{usable}}{I_{battery}} = 0,4414\ hr$

$t_{operating,min} := t_{operating} = 26,4851\ min$ //The theoretical worst-case operating time at the calculated maximum load

Operating Time at 58% Brightness

$P_{in,58} := \frac{P_{total,58}}{\eta_{converter}} = 27,5661\ W$

$P_{loss,58} := P_{in,58} - P_{total,58} = 2,7566\ W$

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$I_{battery,58} := \frac{P_{in,58}}{V_{battery,nom}} = 3,8286\ A$

$C_{rate,58} := \frac{I_{battery,58}\ hr}{C_{battery}} = 1,1261$

$t_{operating,58} := \frac{C_{usable}}{I_{battery,58}} = 0,7104\ hr$

$t_{operating,58,min} := t_{operating,58} = 42,6263\ min$

Operating Time at Measured Load

$V_{battery,meas} := 7,26\ V$ //Measured battery voltage (Table 5.2)

$I_{battery,meas} := 2,47\ A$ //Measured maximum battery current (Table 5.2)

$P_{battery,meas} := V_{battery,meas} \cdot I_{battery,meas} = 17,9322\ W$

$\eta_{measured} := \frac{P_{measured,max}}{P_{battery,meas}} = 0,8114$ //Effective converter efficiency at the measured load

$P_{loss,meas} := P_{battery,meas} - P_{measured,max} = 3,3822\ W$ //Converter heat at the measured load

$C_{rate,meas} := \frac{I_{battery,meas}\ hr}{C_{battery}} = 0,7265$

$t_{operating,meas} := \frac{C_{usable}}{I_{battery,meas}} = 66,0729\ min$

$SF_{BMS,meas} := \frac{I_{BMS,rated}}{I_{battery,meas}} = 5,2632$

Conclusion: Using the measured battery current gives a more realistic operating time of about 66 minutes, compared with 26.5 minutes at the theoretical maximum load and 42.6 minutes at the 58% brightness limit. The measured converter efficiency is about 81%, slightly below the assumed 90%.

2.1.7 Battery Charging (CN3302):

$V_{charge} := 8,4\ V$

$I_{charge} := 2,2\ A$ //CN3302 charging current

$C_{rate,charge} := \frac{I_{charge}\ hr}{C_{battery}} = 0,6471$

$I_{cell,charge,max} := 2\ A$ //Samsung INR18650-35E maximum charge current (datasheet)

$SF_{charge} := \frac{I_{cell,charge,max}}{I_{charge}} = 0,9091$

$t_{charge,CC} := \frac{C_{battery}}{I_{charge}} = 1,5455\ hr$ //Constant-current stage

$k_{CV} := 1,2$ //Allowance for the slower constant-voltage stage

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$t_{charge} := k_{CV} \cdot t_{charge,CC} = 1,8545\ hr$

$P_{charge} := V_{charge} \cdot I_{charge} = 18,48\ W$

$\eta_{charger} := 0,9$

$V_{USB} := 5\ V$

$I_{charger,in} := \frac{P_{charge}}{\left(\eta_{charger} \cdot V_{USB}\right)} = 4,1067\ A$

Conclusion: The 2.2 A charging current corresponds to 0.65 C and gives a full charge in about 1.9 h. It is slightly above the 2 A maximum charge current of the cells, so a lower charging current would be preferable for the final design. A 5 V input would need to supply about 4.1 A for the full charging current.

2.1.8 Wire Sizing:

$rho_{Cu} := \frac{1,68 \cdot 10^{(-8)}\ V}{A}\ m$ //Resistivity of copper

$L_{wire} := 0,5\ m$ //Conservative one-way cable length

$k_{drop} := 0,05$ //Maximum allowable voltage drop of 5%

Main power connections

$V_{drop,power} := k_{drop} \cdot V_{battery,nom} = 0,36\ V$

$A_{power} := \frac{2 \cdot I_{battery} \cdot rho_{Cu} \cdot L_{wire}}{V_{drop,power}} = 0,2876\ mm^2$

$A_{20AWG} := 0,519\ mm^2$

$SF_{wire,power} := \frac{A_{20AWG}}{A_{power}} = 1,8049$

LED strip connections (per strip)

$I_{strip,max} := \frac{I_{LED,total}}{2} = 3,6\ A$

$I_{strip,58} := I_{strip,max} \cdot Brightness = 2,088\ A$

$V_{drop,5V} := k_{drop} \cdot V_5 = 0,25\ V$

$A_{LED} := \frac{2 \cdot I_{strip,max} \cdot rho_{Cu} \cdot L_{wire}}{V_{drop,5V}} = 0,2419\ mm^2$

$A_{LED,58} := \frac{2 \cdot I_{strip,58} \cdot rho_{Cu} \cdot L_{wire}}{V_{drop,5V}} = 0,1403\ mm^2$

//Compare with the cross-sectional area of the LED strip power leads (e.g. 22 AWG = 0.326 mm², 20 AWG = 0.519 mm²).

Microcontroller connections

$V_{drop,3,3V} := k_{drop} \cdot V_{3,3V} = 0,165\ V$

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$$A_{MCU} := \frac{2 \cdot I_{pico} \cdot rho_{Cu} \cdot L_{wire}}{V_{drop,3,3V}} = 0,0102\ mm^2$$

$$A_{22AWG} := 0,326\ mm^2$$

Speaker and amplifier connections

$$P_{speaker} := 3\ W$$

$$R_{speaker} := \frac{4\ V}{A}$$ //4 Ω speaker

$$I_{speaker} := \sqrt{\frac{P_{speaker}}{R_{speaker}}} = 0,866\ A$$

$$A_{speaker} := \frac{2 \cdot I_{speaker} \cdot rho_{Cu} \cdot L_{wire}}{V_{drop,5V}} = 0,0582\ mm^2$$

$$A_{26AWG} := 0,129\ mm^2$$

Conclusion: All selected wires have a larger cross-sectional area than required for a 5% voltage drop. The main power wire (20 AWG) has an area 1.8 times the required value.

2.1.9 Connector and Terminal Current Checks:

$$I_{JST,rated} := 3\ A$$ //JST-XH 2.54 mm rated current

$$SF_{JST,max} := \frac{I_{JST,rated}}{I_{strip,max}} = 0,8333$$ //Full white brightness

$$SF_{JST,58} := \frac{I_{JST,rated}}{I_{strip,58}} = 1,4368$$ //58% brightness limit

$$I_{terminal,rated} := 10\ A$$ //Screw terminal block rating - confirm against the datasheet of the terminals used

$$SF_{terminal} := \frac{I_{terminal,rated}}{I_{battery}} = 1,6229$$

Conclusion: At full white brightness each strip would draw 3.6 A, which exceeds the 3 A JST-XH rating. With the brightness limited to 58% in software the strip current is 2.09 A, giving a safety factor of 1.44. The screw terminals carry the maximum battery current with a safety factor of 1.62.

2.2 Mechanical Design:

2.2.1 Design Assumptions and Physical Dimensions:

Hilt

$$L_{hilt} := 250\ mm \qquad g := 9,81\ \frac{m}{s^2} \qquad rho_{PETG} := 1,27\ \frac{g}{cm^3}$$

$$V_{hilt} := 347,22\ cm^3 \qquad Infill := 0,7$$

$$m_{hilt} := V_{hilt} \cdot rho_{PETG} \cdot Infill = 0,3087\ kg$$

$$W_{hilt} := \left(m_{hilt}\right) \cdot g = 3,0281\ N$$

Blade Geometry

$$L_{blade} := 920\ mm \qquad D_o := 25,4\ mm$$

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$t_{blade} := 3\ mm$ $rho_{PC} := 1,2\ \frac{g}{cm^3}$

$D_i := D_o - 2 \cdot t_{blade} = 19,4\ mm$

$A_{blade} := \left(\frac{\pi}{4}\right) \cdot \left(D_o^2 - D_i^2\right) = 0,0002\ m^2$

$V_{blade} := A_{blade} \cdot L_{blade} = 0,0002\ m^3$

$m_{blade} := V_{blade} \cdot rho_{PC} = 0,2331\ kg$

$W_{blade} := m_{blade} \cdot g = 2,2864\ N$

$w_{blade} := \frac{W_{blade}}{L_{blade}} = 2,4852\ m\ Pa$

Component Loading

$m_1 := 0,14\ kg$ $m_2 := 0,33\ kg$

$W_1 := m_1 \cdot g = 1,3734\ N$ $W_2 := m_2 \cdot g = 3,2373\ N$

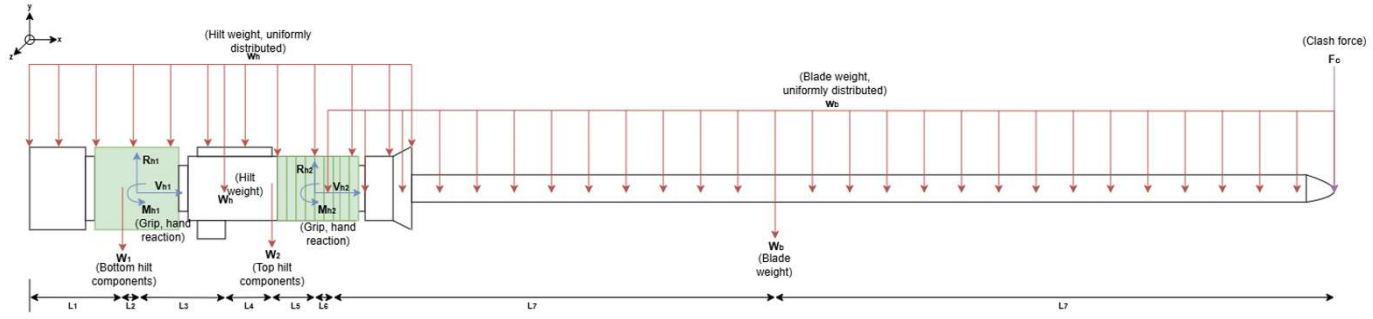
$x_1 := 66,5\ mm$ $x_{hilt} := 125\ mm$ $x_2 := 191\ mm$ $x_{blade} := 710\ mm$

$x_{R1} := 92\ mm$ $x_{R2} := 242,5\ mm$

2.2.2 Static Equilibrium of the Complete Lightsabre:

Free Body Diagram:

//Blade is modelled as a cantilever beam fixed at the blade mount.



$\Sigma M_{R1} := 0$

$\Sigma M_{R1} := \left(R_{h2} \cdot \left(x_{R2} - x_{R1}\right) + W_1 \cdot \left(x_{R1} - x_1\right) - W_{hilt} \cdot \left(x_{hilt} - x_{R1}\right) - W_2 \cdot \left(x_2 - x_{R1}\right) - W_{blade} \cdot \left(x_{blade} - x_{R1}\right)\right)$

$R_{h2} := \frac{\left(W_{hilt} \cdot \left(x_{hilt} - x_{R1}\right) + W_2 \cdot \left(x_2 - x_{R1}\right) + W_{blade} \cdot \left(x_{blade} - x_{R1}\right) - W_1 \cdot \left(x_{R1} - x_1\right)\right)}{\left(x_{R2} - x_{R1}\right)} = 11,9496\ N$

$\Sigma F_y := 0$

$\Sigma F_y := \left(R_{h1} + R_{h2} + - W_1 - W_{hilt} - W_2 - W_{blade}\right)$

$R_{h1} := W_1 + W_{hilt} + W_2 + W_{blade} - R_{h2} = -2,0243\ N$

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//R.h1 is negative, so the rear hand pushes downward on the hilt while the front hand carries the full weight plus the moment from the overhanging blade.

Selection of the Clash Load

$F_c := 50\ \mathrm{N}$

$m_c := \frac{F_c}{g} = 5,0968\ \mathrm{kg}$

$F_{ratio} := \frac{F_c}{W_{blade}} = 21,8682$

Conclusion: The clash force was not measured experimentally, so a 50 N transverse force was assumed for the structural analysis. This represents a severe impact condition and is approximately 21.8 times the blade weight, providing a conservative basis for the stress and deflection calculations.

2.2.3 Blade Loading Cases:

//The blade is assessed under two loading conditions. Static loading represents the blade's own weight, while the clash case represents a transverse impact at the blade tip. The clash case is used as the main structural design condition.

Static loading

$V_{static} := w_{blade} \cdot L_{blade} = 2,2864\ \mathrm{N}$

$M_{static} := w_{blade} \cdot \frac{L_{blade}^{2}}{2} = 1,0518\ \mathrm{N\ m}$

Clash loading

//Clash force acts transversely at the blade tip.

$V_{clash} := F_c + W_{blade} = 52,2864\ \mathrm{N}$

$M_{clash} := F_c \cdot L_{blade} + M_{static} = 47,0518\ \mathrm{N\ m}$

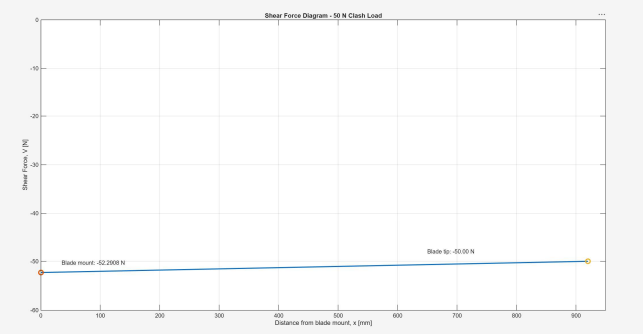
2.2.4 Shear Force Diagram:

//x = 0 at the blade mount and x = L_blade at the blade tip.

$V(x) := -\left(F_c + w_{blade} \cdot \left(L_{blade} - x\right)\right)$

$V_{mount} := V(0) = -52,2864\ \mathrm{N}$

$V_{tip} := V\left(L_{blade}\right) = -50\ \mathrm{N}$

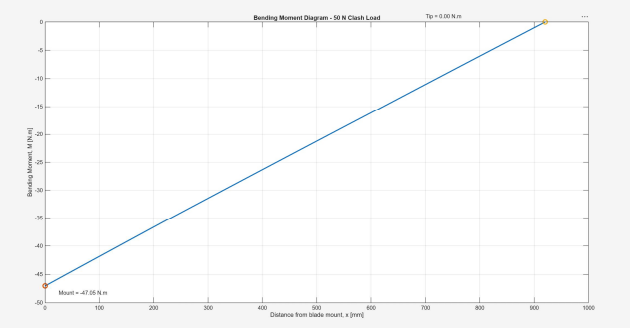


2.2.5 Bending Moment Diagram:

$M(x) := -\left(F_c \cdot \left(L_{blade} - x\right) + w_{blade} \cdot \frac{\left(L_{blade} - x\right)^{2}}{2}\right)$

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$M_{mount} := M(0) = -47,0518\ \text{N m}$

$M_{tip} := M\left(L_{blade}\right) = 0\ \text{N m}$

2.2.6 Bending Stress:

// Maximum bending stress occurs at the blade mount. Linear elastic beam theory is used.

$M_{max} := \left|M_{mount}\right| = 47,0518\ \text{N m}$

$I := \frac{\pi}{64} \cdot \left(D_o^{\,4} - D_i^{\,4}\right) = 1,3479 \cdot 10^{-8}\ \text{m}^4$

$c := \frac{D_o}{2} = 0,0127\ \text{m}$

$\sigma_{max} := \frac{M_{max} \cdot c}{I} = 44,3337\ \text{MPa}$

$\sigma_{static} := \frac{M_{static} \cdot c}{I} = 0,991\ \text{MPa}$ //Bending stress under the blade self-weight only

2.2.7 Transverse Shear Stress:

$V_{max} := \left|V_{mount}\right| = 52,2864\ \text{N}$

$R_o := \frac{D_o}{2} = 12,7\ \text{mm}$ $R_i := \frac{D_i}{2} = 9,7\ \text{mm}$

$Q := \left(\frac{2}{3}\right) \cdot \left(R_o^{\,3} - R_i^{\,3}\right) = 7,5714 \cdot 10^{-7}\ \text{m}^3$ // First moment of area at the neutral axis

$b := 2 \cdot \left(R_o - R_i\right) = 6\ \text{mm}$ // Width of hollow section at neutral axis

$\tau_{max} := V_{max} \cdot \frac{Q}{\left(I \cdot b\right)} = 0,4895\ \text{MPa}$

2.2.8 Stress Safety Factor:

// A preliminary polycarbonate yield strength of 60 MPa is used for the structural assessment.

$\sigma_y := 60\ \text{MPa}$

$FS := \frac{\sigma_y}{\sigma_{max}} = 1,3534$

2.2.9 Blade Deflection:

// Elastic modulus is assumed as 2.3 GPa for the polycarbonate.

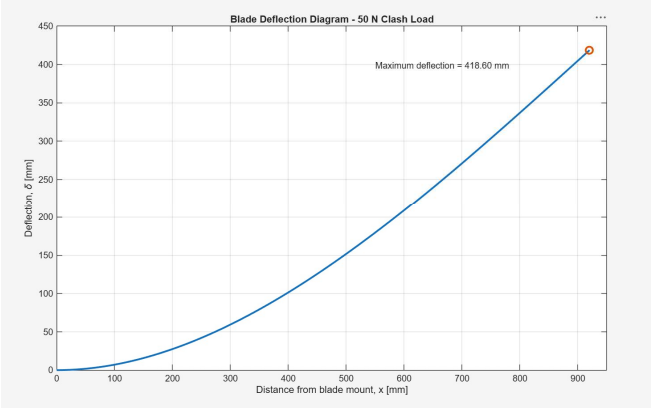
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$E := 2,3\ \text{GPa}$

$\delta_{clash} := F_c \cdot \frac{L_{blade}^{3}}{(3 \cdot E \cdot I)} = 418,6377\ \text{mm}$

$\delta_{ratio} := \frac{\delta_{clash}}{L_{blade}} = 0,455$



$\delta_{static} := \frac{w_{blade} \cdot L_{blade}^{4}}{(8 \cdot E \cdot I)} = 7,1789\ \text{mm}$

//Tip deflection under the blade self-weight

$\delta_{total} := \delta_{clash} + \delta_{static} = 425,8166\ \text{mm}$

Conclusion: The calculated tip deflection under the clash load is about 45% of the blade length, which is outside the small-deflection range for which the cantilever formula is valid, so the value is only indicative. It shows that a 50 N static tip load would bend the blade severely. In practice a clash is a short impact rather than a sustained static load, but blade stiffness (wall thickness or material) should be reviewed.

2.2.10 Blade Mount Epoxy Bond:

//Blade engagement inside the mount is taken as 80 mm. Epoxy is assumed to form a continuous cylindrical bond around the blade over the 80 mm engagement.

$D_{blade} := 0,0254\ \text{m}$

$L_{mount} := 0,08\ \text{m}$

$R_{blade} := \frac{D_{blade}}{2} = 0,0127\ \text{m}$

$A_{bond} := \pi \cdot D_{blade} \cdot L_{mount} = 0,0064\ \text{m}^{2}$

Direct shear in epoxy

$\tau_{direct} := \left| \frac{V_{mount}}{A_{bond}} \right| = 0,0082\ \text{MPa}$

$M_{mount_load} := \left| M_{mount} \right| = 47,0518\ \text{N m}$

$\tau_{moment} := \frac{M_{mount_load}}{\pi \cdot R_{blade}^{2} \cdot L_{mount}} = 1,1607\ \text{MPa}$

$\tau_{epoxy} := \sqrt{\tau_{direct}^{2} + \tau_{moment}^{2}} = 1,1608\ \text{MPa}$

Epoxy Strength

// Bostik Super Epoxy TDS (2025) gives a bulk tensile strength of 50 MPa. A conservative allowable bond stress of 2 MPa is

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$\tau_{allowable} := 2\ MPa$ used because adhesion to PETG and polycarbonate governs the joint strength.

$FS_{epoxy} := \frac{\tau_{allowable}}{\tau_{epoxy}} = 1,723$

Blade Mount Flange (Lip) Bond to the Hilt

//The PETG blade mount sits on the top of the hilt and is bonded with epoxy on the lip (flange) at its base. The bond is modelled as an annulus between the Ø25 mm mount bore and the Ø53 mm top of the hilt, carrying the clash moment and shear at the mount.

$D_{lip,o} := 53\ mm$

$D_{lip,i} := 25\ mm$

$A_{lip} := \frac{\pi}{4} \cdot \left(D_{lip,o}^{2} - D_{lip,i}^{2} \right) = 1715,3096\ mm^{2}$

$I_{lip} := \frac{\pi}{64} \cdot \left(D_{lip,o}^{4} - D_{lip,i}^{4} \right) = 3,6815 \cdot 10^{5}\ mm^{4}$

$\sigma_{lip} := \frac{M_{mount_load} \cdot \left(\frac{D_{lip,o}}{2} \right)}{I_{lip}} = 3,3869\ MPa$ //Tensile (peel) stress at the edge of the bond

$\tau_{lip} := \frac{V_{max}}{A_{lip}} = 0,0305\ MPa$

$FS_{lip} := \frac{\tau_{allowable}}{\sigma_{lip}} = 0,5905$ //Using the 2 MPa tensile strength of the epoxy

$F_{c,allow} := \frac{\left(\frac{\tau_{allowable} \cdot I_{lip}}{\left(\frac{D_{lip,o}}{2} \right)} - M_{static} \right)}{L_{blade}} = 29,0576\ N$ //Tip force at which the lip bond reaches its strength

Conclusion: The lip bond has a factor of safety of 0.59 under the assumed 50 N clash load, so it is the weakest part of the blade connection. The bond reaches its strength at a tip force of about 29 N. Light clashes are therefore acceptable, but heavy duelling impacts could separate the blade mount from the hilt. A longer spigot into the hilt or a mechanical fastener would increase the strength.

2.3 Thermal Design:

2.3.1 LED Heat Dissipation:

$\eta_{LED,light} := 0,2$ //Assumed fraction of LED power converted to light

$P_{LED,meas} := P_{measured,max} - \left(P_{amp} + P_{pico} + P_{encoder} + P_{3,3V} \right) = 10,6205\ W$ //Approximate LED power at the measured load

$Q_{LED,58} := \left(1 - \eta_{LED,light} \right) \cdot P_{LED,58} = 16,704\ W$

$Q_{LED,meas} := \left(1 - \eta_{LED,light} \right) \cdot P_{LED,meas} = 8,4964\ W$

$A_{blade,surf} := \pi \cdot D_{o} \cdot L_{blade} = 0,0734\ m^{2}$

$h_{conv} := \frac{10\ W}{\left(m^{2}\ K \right)}$ //Natural convection heat transfer coefficient for still air

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$$\Delta T_{blade,58} := \frac{Q_{LED,58}}{\left(h_{conv} \cdot A_{blade,surf}\right)} = 22,7535\ K$$

$$\Delta T_{blade,meas} := \frac{Q_{LED,meas}}{\left(h_{conv} \cdot A_{blade,surf}\right)} = 11,5735\ K$$

Conclusion: The blade surface temperature rise is about 23 K at the 58% brightness limit and about 12 K at the measured load. With a 25 °C ambient this gives a maximum of about 48 °C, below the 80 °C maximum operating temperature of the WS2812B, so the aluminium heat-sink rod is not required.

2.3.2 Hilt Vents:

$$N_{vent} := 2$$

$$w_{vent} := 2\ mm$$

$$L_{vent} := 37\ mm$$ //Two through-slots in the top section of the hilt

$$A_{vent} := N_{vent} \cdot w_{vent} \cdot L_{vent} = 148\ mm^2$$

$$Q_{conv} := P_{loss,meas} = 3,3822\ W$$ //Heat produced by the buck converter at the measured load

$$rho_{air} := \frac{1,2\ kg}{3\ m}$$

$$cp_{air} := \frac{1005\ J}{(kg\ K)}$$

$$v_{air} := \frac{0,2\ m}{s}$$ //Assumed natural convection air velocity through the vents

$$\Delta T_{air} := 10\ K$$ //Allowed air temperature rise

$$Q_{vent} := rho_{air} \cdot cp_{air} \cdot v_{air} \cdot A_{vent} \cdot \Delta T_{air} = 0,357\ W$$

$$Q_{vent,fraction} := \frac{Q_{vent}}{Q_{conv}} = 0,1055$$

Remaining heat through the hilt wall

$$D_{hilt} := 53\ mm$$

$$t_{wall} := 3\ mm$$

$$k_{PETG} := \frac{0,2\ W}{(m\ K)}$$

$$A_{hilt} := \pi \cdot D_{hilt} \cdot L_{hilt} = 0,0416\ m^2$$

$$R_{wall} := \frac{t_{wall}}{\left(k_{PETG} \cdot A_{hilt}\right)} = 0,3604\ \frac{K}{W}$$

$$R_{surf} := \frac{1}{\left(h_{conv} \cdot A_{hilt}\right)} = 2,4023\ \frac{K}{W}$$

$$\Delta T_{hilt} := \left(Q_{conv} - Q_{vent}\right) \cdot \left(R_{wall} + R_{surf}\right) = 8,3578\ K$$

Conclusion: The two vents remove about 0.36 W, roughly 11% of the 3.4 W produced by the converter at the measured load. The remaining heat passes through the hilt wall, giving an estimated internal temperature rise of about 8 K above ambient. The vents therefore assist the cooling, while the heatsink and hilt surface carry most of the heat. The measured temperatures in

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Table 5.3 verify this.

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